From Attribute Alignment to Attribute Verification: GARNET for Text-Based Person Retrieval

Supplementary Material

The main paper identifies a specific gap in text-based person retrieval (TBPR): prior work has substantially advanced global image–text matching, fine-grained/attribute correspondence, and perturbation sensitivity. The remaining limitation is not that prior methods ignore attributes, local alignment, or textual perturbations; rather, these capabilities do not necessarily impose the specific requirement that local visual evidence for the *exact changed phrase* decrease after a controlled semantic corruption. GARNET addresses this missing coupling through Phrase–Patch Grounding (PPG) and Counterfactual Attribute Verification (CAV). This supplement provides additional quantitative evidence, controlled-evaluation details, implementation and paraphrase-generation details, and qualitative analyses of the same model and evaluation setting as the main paper.

A. Extended Quantitative Analysis

A.1. Changed-Phrase Specificity

CAV-L should selectively change the evidence of the deliberately corrupted phrase rather than destabilize every phrase in the caption. For a counterfactual changing phrase index c_{ik}, we measure unchanged-phrase drift as

$$\delta_{\text{unch}} = \frac{1}{NK} \sum_{i,k} \frac{1}{P_i - 1} \sum_{\substack{r=1 \\ r \neq c_{ik}}}^{P_i} \left| a(I_i, p_{ir}) - a(I_i, \widetilde{p}_{ikr}) \right|, \tag{1}$$

where $\widetilde{p}_{ikr}$ is the corresponding unchanged phrase encoded in the counterfactual caption. Samples with fewer than two valid phrases are excluded.

Table 1. **Changed-phrase specificity.** A useful verifier should increase the changed-phrase gap while reducing unrelated phrase drift.

Variant	$\Delta_\ell \uparrow$	$\delta_{\text{unch}} \downarrow$
PPG only	0.106	0.044
PPG + CAV-G	0.115	0.040
PPG + CAV-G + CAV-L	**0.135**	**0.035**

From PPG to the full model, the changed-phrase gap increases from 0.106 to 0.135, while unrelated drift decreases from 0.044 to 0.035. Relative to PPG+CAV-G, CAV-L increases Δ_ℓ by 0.020 and reduces δ_{unch} by 0.005, indicating that the stronger local separation is concentrated on the semantic content actually corrupted.

Table 2. **Local C>C (%) by corruption type.**

Variant	Color	Clothing	Object	Pose/Action
PPG only	96.79	83.27	90.48	83.05
PPG + CAV-G	**98.72**	84.01	93.12	86.44
Full GARNET	98.08	**86.25**	**94.71**	**90.68**

A.2. Corruption-Type Analysis

Color is already close to saturation with PPG+CAV-G (98.72%), so adding CAV-L produces a small decrease to 98.08%. The harder categories improve: clothing rises from 84.01% to 86.25%, carried objects from 93.12% to 94.71%, and pose/action from 86.44% to 90.68%. This non-uniform pattern avoids claiming an artificial improvement in every category while showing that CAV-L is particularly useful when verification requires richer visual context.

B. Controlled Evaluation Protocol

The controlled evaluation directly tests the failure mode motivating GARNET. A model may achieve strong global compatibility, local correspondence, or perturbation sensitivity without explicitly requiring the visual support of the *specific corrupted phrase* to decrease. The following protocols isolate that behavior.

B.1. Attribute-Confusable Gallery Construction

Attr-Conf. R@1 evaluates retrieval when the gallery contains different-identity near misses that preserve much of the query semantics but conflict in at least one visually verifiable attribute. Same-identity candidates are first removed. We then retain captions with moderate-to-high lexical overlap and require an explicit conflict in color, clothing, carried object, or pose/action. Eligible candidates are ranked by cosine similarity using a frozen, method-independent `all-mpnet-base-v2` Sentence-BERT encoder [4]; the top-8 hard negatives are kept. A light method-independent sanity check removes annotation inconsistencies, non-visible queried attributes, and accidental identity leakage.

For query T_i and candidate T_j, lexical overlap is measured by

$$J(T_i, T_j) = \frac{|\mathcal{W}_i \cap \mathcal{W}_j|}{|\mathcal{W}_i \cup \mathcal{W}_j|}, \tag{2}$$

where $\mathcal{W}_i$ and $\mathcal{W}_j$ are stopword-filtered content-word sets.

B.2. Post-hoc Local Probe for Prior Methods

IRRA [3], RaSa [1], MARS++ [2], and CECA [6] do not natively define GARNET's phrase-to-patch support score.

Table 3. **Attribute-confusable gallery construction.**

Setting	Value
Same-identity candidates	Removed
Jaccard overlap	$[0.6, 0.95]$
Conflict categories	Color / Clothing / Object / Pose
Candidate-pool cap	100
Independent ranking encoder	`all-mpnet-base-v2`
Hard negatives per query	8
Insufficient-candidate queries	Excluded
Final sanity check	Method-independent

Their Local C>C and Δ_ℓ values therefore use a common *evaluation-only* probe on frozen checkpoints. No baseline is retrained and no baseline architecture is modified. All methods are evaluated on the same controlled counterfactual set.

For every valid changed phrase, local support is computed as

$$a(I, p) = \max_m \mathbf{u}_p^\top \mathbf{v}_m, \tag{3}$$

where $\mathbf{u}_p$ is the normalized phrase representation and $\mathbf{v}_m$ is the normalized representation of visual patch m.

RaSa and MARS++. Both use an ALBEF-style ViT/BERT pipeline. The visual CLS token is removed and the remaining patch tokens are passed through each method's existing `vision_proj`. The changed phrase is localized using the method's native BERT WordPiece tokenizer after its standard caption preprocessing; hidden states over the exact phrase span are mean-pooled and passed through the existing `text_proj`.

IRRA and CECA. Both use CLIP-family image/text representations. The visual CLS token is removed, the exact changed phrase is mapped to its contiguous CLIP-BPE span, and token states in that span are mean-pooled. Their encoder outputs already lie in the shared retrieval space, so no new projection head is introduced.

For both tokenizer families, samples whose phrase is truncated or cannot be localized exactly are excluded from both numerator and denominator. GARNET is different in an important respect: its local score is the *native PPG support learned during training*, whereas prior-method local scores are post-hoc diagnostic measurements. The probe therefore asks whether the existing representation of each prior method already provides local evidence that rejects the exact corrupted phrase.

C. Implementation and Reproducibility Details

We use CLIP ViT-L/14 as the dual-encoder backbone. Stage A performs person-domain pre-training on MALS [5]; PPG and CAV are introduced during benchmark-specific Stage B fine-tuning so that gains from representation strengthening remain separable from the proposed grounded verification mechanism. Table 5 summarizes the principal architecture and optimization settings. We use cosine learning-rate decay with warm-up, gradient clipping at 1.0, and paraphrase substitution with probability 0.5 during Stage B. The cross-modal MLM branch receives a separate masked copy of the correct caption so that reconstruction does not directly counteract the CAV corruption signal. Headline evaluation uses normalized EMA global features without local re-ranking.

Table 4. **Fixed lexical substitutions used for rule-based paraphrasing.**

Original	Replacement(s)
pedestrian	person, individual
person	pedestrian, individual
individual	person, pedestrian
man	male, guy
woman	female, lady
wearing	dressed in, clad in
carrying	holding
holding	carrying
walking	strolling, moving
shoes	footwear
pants	trousers
trousers	pants
bag	satchel
hair	hairstyle
shirt	top
jacket	coat
young	youthful
old	elderly, aged
tall	lanky
short	small in stature
black	dark
white	light-colored

Phrase and counterfactual preprocessing. For each caption T_i, visually verifiable phrases are extracted and their token spans are stored. Each accepted counterfactual $\widetilde{T}_{ik}$ changes exactly one recorded phrase while preserving the remainder of the description. We use four corruption categories:

Color, Clothing, Carried Object, Pose/Action.

The preprocessing record stores the changed phrase index c_{ik}, its original phrase $p_{i,c_{ik}}$, the corrupted phrase $\widetilde{p}_{i,c_{ik}}$, and the corruption category. Only examples with valid tokenizer spans and visually verifiable changes are retained. This controlled construction is crucial to GARNET because CAV-L can supervise the local evidence of the *known changed phrase*, rather than treating the entire corrupted caption as an undifferentiated hard negative.

Table 5. **GARNET implementation settings.**

Setting	Stage A	Stage B
Backbone	CLIP ViT-L/14	CLIP ViT-L/14, initialized from Stage A
Input resolution	336×112	336×112
Patch grid	24×8 ($N_p = 192$)	24×8 ($N_p = 192$)
Effective batch size	256	128
Optimizer	Adam	Adam
Backbone / new-layer LR	10^{-5} / 5×10^{-5}	10^{-5} / 5×10^{-5}
Weight decay	4×10^{-5}	4×10^{-5}
Training epochs	5	60 (CUHK/ICFG), 30 (RSTPReid)
CAV margins (m_g, m_ℓ)	–	$(0.10, 0.10)$
Counterfactuals per caption K	–	3
Loss weights	SDM=1, MLM=1	SDM=1, PPG=0.5, CAV=0.5, MLM=1, ID=1
EMA momentum	–	0.999

D. Paraphrase Generation and Semantic Preservation

GARNET uses the same deterministic rule-based paraphrase generator for training-time augmentation and the semantic-preservation diagnostic. **No LLM or external generative API is used.** The procedure applies (i) seeded lexical substitution from a fixed hand-crafted dictionary, with at most two substitutions per variant, and (ii) clause reordering with probability 0.5 when multiple period-delimited clauses are available. Up to two distinct variants are retained; unchanged or duplicate outputs are discarded. Generation is deterministic per dataset/caption index using `random.Random`.

Because several mappings are contextual lexical replacements rather than strict dictionary synonyms in every sentence, we refer to the resource as a *fixed lexical substitution dictionary*. Table 4 reports the rules used in the reported experiments.

The training caches contain 68,126 caption records for CUHK-PEDES, 34,674 for ICFG-PEDES, and 37,010 for RSTPReid. In the X-C4 diagnostic, 6,148 of 6,156 captions receive at least one accepted paraphrase (99.87%).

Table 6. **Paraphrase robustness on CUHK-PEDES.** Δ_{para} denotes the R@1 drop from the original to the meaning-preserving paraphrased query.

Method	Original R@1 ↑	Paraphrase R@1 ↑	Δ_{para} ↓
IRRA [3]	73.15	69.93	3.22
RaSa [1]	76.38	72.77	3.61
GARNET	**79.15**	**79.18**	**-0.03**

The near-zero change for GARNET is consistent with robustness to meaning-preserving lexical rewording. This complements the counterfactual test: surface-form changes should preserve compatibility, while a controlled semantic contradiction should reduce global and local support.

E. Additional Qualitative Analysis

The following visualizations complement the quantitative diagnostics without changing the main-paper model or evaluation protocol. Together they separate four questions: whether controlled corruption changes local evidence, whether that change affects retrieval behavior while preserving paraphrases, what CAV-L contributes beyond PPG+CAV-G, and where the full model still fails.

E.1. Prior Work vs. GARNET under Controlled Semantic Corruption

Figure 1 compares gradient-activation responses for the original phrase and a controlled semantic corruption. The examples cover color (*red shirt→tan shirt*) and object (*bag→cup*) changes. In these representative examples, RaSa and MARS++ retain broadly similar response patterns after corruption, whereas GARNET shows a stronger change in the evidence associated with the changed phrase. These maps are an evaluation-only visualization and are distinct from GARNET's native PPG support map. A fixed visualization scale is used within each architecture, so original-versus-corrupted changes should be interpreted primarily *within* a method rather than as absolute activation comparisons across methods.

E.2. Paraphrase Preservation vs. Semantic Corruption

Figure 2 contrasts a meaning-preserving hard-rule paraphrase with a controlled color corruption. The paraphrase changes only *woman→female* and *pants→trousers*, introducing no new visual attribute. For the original and paraphrased descriptions, the paired identity remains highly ranked. After changing the specified attribute (*gray pants→white pants*), IRRA and RaSa still retain the paired identity at Rank-1 in this representative example, whereas GARNET demotes it to Rank-2. This behavior illustrates the distinction between robustness to surface-form rewording and sensitivity to a semantic contradiction.

E.3. Representative Retrieval Improvements after Adding CAV-L

The core ablation shows that adding CAV-L improves exact changed-phrase verification. Figure 3 provides two represen-

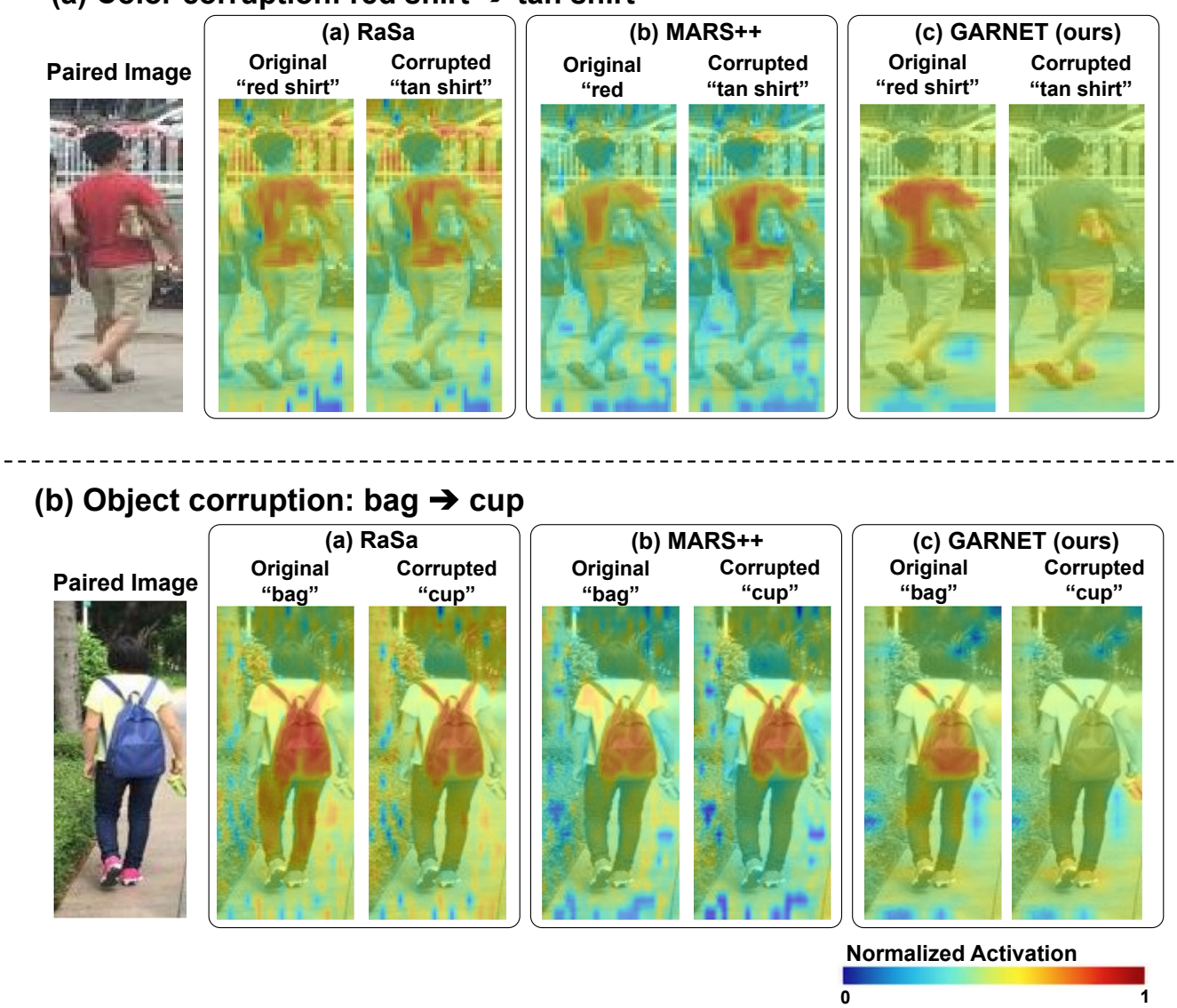


Figure 1. **Prior work vs. GARNET under controlled semantic corruption.** Gradient activation is shown for original and corrupted phrases using the same fixed scale within each architecture. Blue denotes low normalized activation (0) and red denotes high normalized activation (1). In these representative examples, RaSa and MARS++ retain relatively similar activation patterns after the semantic change, whereas GARNET exhibits a stronger response change. This evaluation-only visualization is distinct from GARNET's native PPG support map.

tative retrieval examples in which PPG+CAV-G places the paired identity at Rank-2, while the full GARNET configuration (+CAV-L) ranks it first. The visualization is intended as qualitative support for the ablation rather than evidence that CAV-L corrects every retrieval failure.

E.4. Representative Failure Cases

Finally, Figure 4 reports residual limitations of the full GARNET model. We select genuine retrieval failures rather than cases caused by an obviously inconsistent ground-truth annotation. The examples cover severe image degradation, partial visibility, reliance on subtle fine-grained cues, and highly attribute-confusable identities.

Overall, the supplementary evidence supports the main-paper interpretation: strong global matching, fine-grained/local correspondence, or generic perturbation sensitivity does not by itself impose the specific requirement that visual evidence for the *exact changed phrase* decrease after a controlled semantic corruption. GARNET targets this missing coupling through PPG and CAV, while the reported

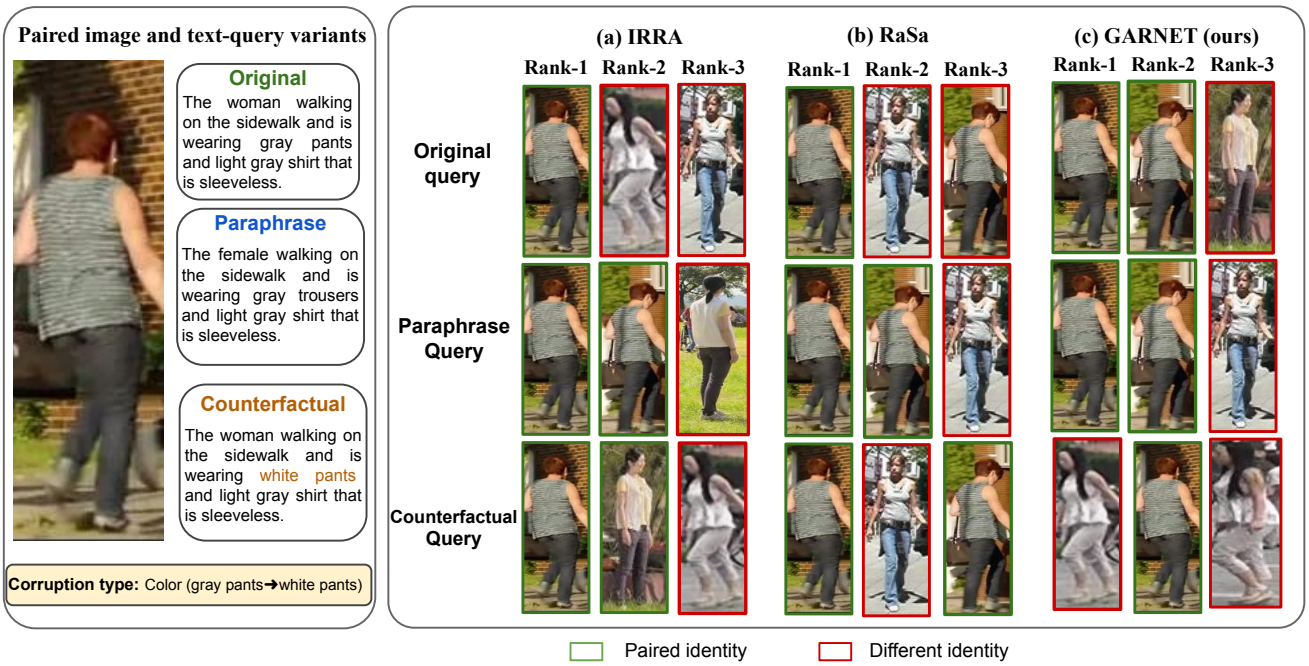


Figure 2. **Paraphrase preservation versus semantic corruption.** Rows correspond to the original text query, a meaning-preserving hard-rule paraphrase, and a controlled counterfactual; columns show Rank-1 to Rank-3 retrievals for IRRA, RaSa, and GARNET. The paraphrase changes only *woman→female* and *pants→trousers*. Green borders denote the paired identity and red borders denote different identities. The representative example shows GARNET preserving the paired identity under paraphrasing while demoting it after semantic attribute corruption.

failure cases identify conditions that remain challenging.

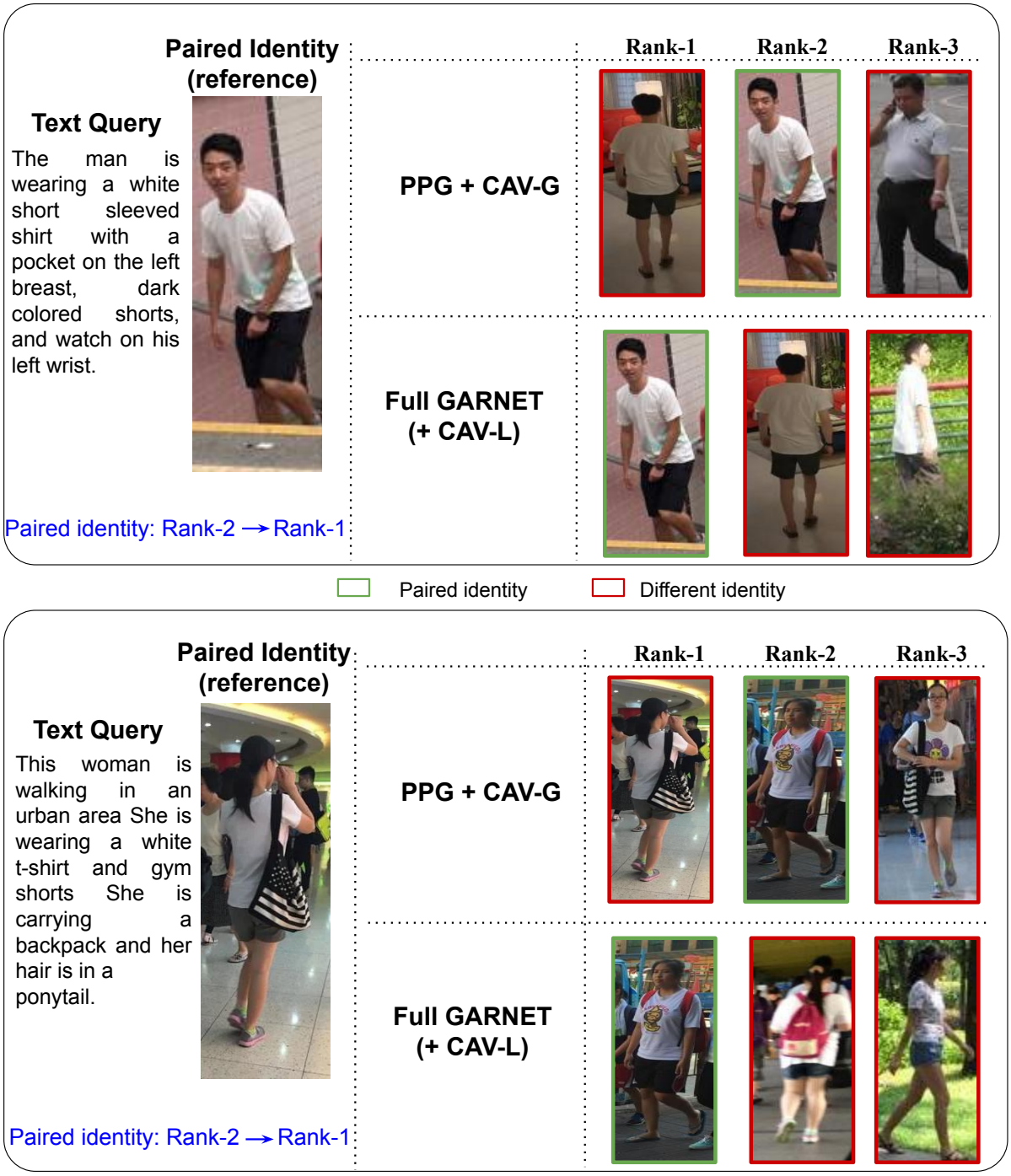


Figure 3. **Representative retrieval improvements after adding CAV-L.** For each text query, the paired identity is shown as a reference image and the top-3 retrievals are compared between PPG+CAV-G and full GARNET (+CAV-L). In both representative examples, the paired identity is ranked second without CAV-L and first in the full configuration. Green borders denote the paired identity and red borders denote different identities.



Figure 4. **Representative failure cases of GARNET.** The four panels illustrate low resolution/blur, partial occlusion, fine-grained attribute ambiguity, and attribute-confusable appearance. Rank-1 to Rank-3 retrievals are shown together with the paired identity at its actual rank. Green borders denote the paired identity and red borders denote different identities.

References

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